5-2c The Variety of Supply Curves

Because the price elasticity of supply measures the responsiveness of quantity supplied to changes in price, it is reflected in the appearance of the supply curve. Figure 5 shows five cases. In the extreme case of zero elasticity, as shown in panel (a), supply is *perfectly inelastic* and the supply curve is vertical. In this case, the quantity supplied is the same regardless of the price. As the elasticity rises, the supply curve gets flatter, which shows that the quantity supplied responds more to changes in the price. At the opposite extreme, shown in panel (e), supply is *perfectly elastic*. This occurs as the price elasticity of supply approaches infinity and the supply curve becomes horizontal, meaning that very small changes in the price lead to very large changes in the quantity supplied.

Figure 5

The Price Elasticity of Supply

The price elasticity of supply determines whether the supply curve is steep or flat. Note that all percentage changes are calculated using the midpoint method.

In some markets, the elasticity of supply is not constant but varies over the supply curve. Figure 6 shows a typical case for an industry in which firms have factories with a limited capacity for production. For low levels of quantity supplied, the elasticity of supply is high, indicating that firms respond substantially to changes in the price. In this region of the supply curve, firms have additional capacity for production, such as plants and equipment that are idle for all or part of the day. Small increases in price make it profitable for firms to begin using this idle capacity. As the quantity supplied rises, firms begin to reach capacity. Once capacity is fully used, further increases in production require the construction of new plants. To induce firms to incur this extra expense, the price must rise substantially, so supply becomes less elastic.

Figure 6

How the Price Elasticity of Supply Can Vary

Because firms often have a maximum capacity for production, the elasticity of supply may be very high at low levels of quantity supplied and very low at high levels of quantity supplied. Here an increase in price from $\$3$ to $\$4$ increases the quantity supplied from 100 to 200. Because the 67 percent increase in quantity supplied (computed using the midpoint method) is larger than the 29 percent increase in price, the supply curve is elastic in this range. By contrast, when the price rises from $\$12$ to $\$15$, the quantity supplied rises only from 500 to 525. Because the 5 percent increase in quantity supplied is smaller than the 22 percent increase in price, the supply curve is inelastic in this range.

Figure 6 presents a numerical example of this phenomenon. When the price rises from $\$3$ to $\$4$ (a 29 percent increase, according to the midpoint method), the quantity supplied rises from 100 to 200 (a 67 percent increase). Because quantity supplied changes proportionately more than the price, the supply curve has an elasticity greater than 1. By contrast, when the price rises from $\$12$ to $\$15$ (a 22 percent increase), the quantity supplied rises from 500 to 525 (a 5 percent increase). In this case, quantity supplied moves proportionately less than the price, so the elasticity is less than 1.

QuickQuiz

5. The price of a good rises from $\$16$ to $\$24$, and the quantity supplied rises from 90 to 110 units. Calculated with the midpoint method, the price elasticity of supply is

   a. $1/5$.

   b. $1/2$.

   c. 2.

   d. 5.

6. If the price elasticity of supply is zero, the supply curve is

a. upward sloping.

b. horizontal.

c. vertical.

d. fairly flat at low quantities but steeper at larger quantities.

7. The ability of firms to enter and exit a market over time means that, in the long run,

a. the demand curve is more elastic.

b. the demand curve is less elastic.

c. the supply curve is more elastic.

d. the supply curve is less elastic.